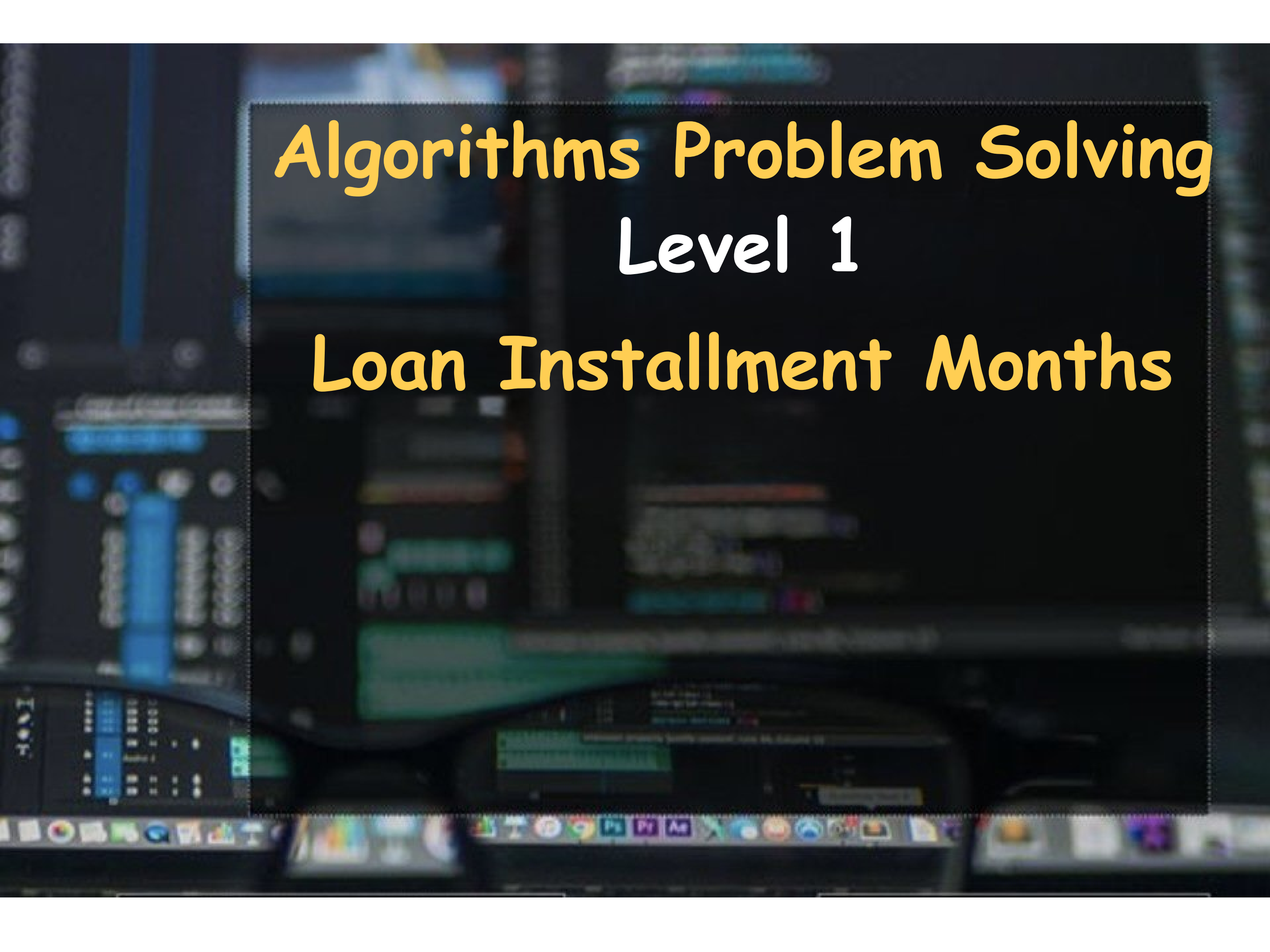




Algorithms Problem Solving Level 1

Loan Installment Months

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower portion of the image, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 1

Loan Installment Months

🔍 Definition

◆ This program reads a loan amount and a monthly payment, then calculates how many months are needed to pay off the loan.

◆ The calculation is:

$\text{LoanAmount} \div \text{MonthlyPayment}$

◆ The program uses a structure to group the loan information and functions to handle input, calculation, and output.

Explanation

- ◆ ``LoanInfo`` is a structure that stores:
 - `LoanAmount`
 - `MonthlyPayment`
- ◆ ``ReadInputs()`` asks the user to enter the loan amount and monthly payment, then returns them as a ``LoanInfo`` object.
- ◆ ``CalculateTotalMonths()`` divides the loan amount by the monthly installment to calculate the number of months.
- ◆ ``PrintResult()`` displays the calculated number of months.
- ◆ In ``main()``, the values stored in the structure are passed to the calculation function, and its result is passed to ``PrintResult()``.

Steps in main()

1. Create the `LoanInfo` structure.
2. Read the loan amount.
3. Read the monthly payment.
4. Return the loan information.
5. Divide the loan amount by the monthly payment.
6. Return the number of months.
7. Print the result.

💡 Example

◆ If the user enters:

Loan Amount: 5000

Monthly Payment: 500

◆ The calculation is:

$$5000 \div 500 = 10$$

➡ Result

You will need 10 months to pay off the loan.

Note

- ◆ The program assumes that the loan is paid with the same fixed amount every month.
- ◆ If the division produces a decimal value, the result can contain a fraction of a month.

⊘ Warning

- ◆ The program does not validate the monthly payment. If `MonthlyPayment` is `0`, division by zero occurs.
- ◆ If the goal is to calculate complete payment months for a real loan, the result would normally need to be rounded up when there is a remaining balance.



Comparison

Exact Division

- ✓ The loan amount divides evenly by the monthly payment.
- ✓ Example: $5000 \div 500 = 10$.
- ✓ The result is a whole number of months.

Non-Exact Division

- ✓ The loan amount does not divide evenly.
- ✓ Example: $5000 \div 600 = 8.3333...$.
- ✓ A real payment schedule would require a final partial payment, so complete months may need to be rounded up.

✦✦ Summary

- ✓ A structure can group the loan amount and monthly payment.
- ✓ The number of months is calculated by dividing the loan amount by the monthly payment.
- ✓ `double` allows the result to contain decimal values.
- ✓ Monthly payment should not be zero.
- ✓ For complete payment months, a non-integer result may need to be rounded up.

Thank you 😊